UM
HACKATHON
2026

KUDOSCAN BOT

SECURITY THAT THINKS BEFORE YOU CLICK.

PRESENTED BY: GROUP NO. 269 (KUDOS)

PROBLEM STATEMENT

Users cannot identify phishing links without technical knowledge. Phishing attacks target everyone which from banking scams to fake deliveries.

One wrong click can cause stolen credentials, financial loss, or malware. This will cause vulnerable groups have the highest risk and the least protection.

OUR OBJECTIVES

INSTANT VERIFICATION

Check a suspicious link in seconds, directly within Telegram

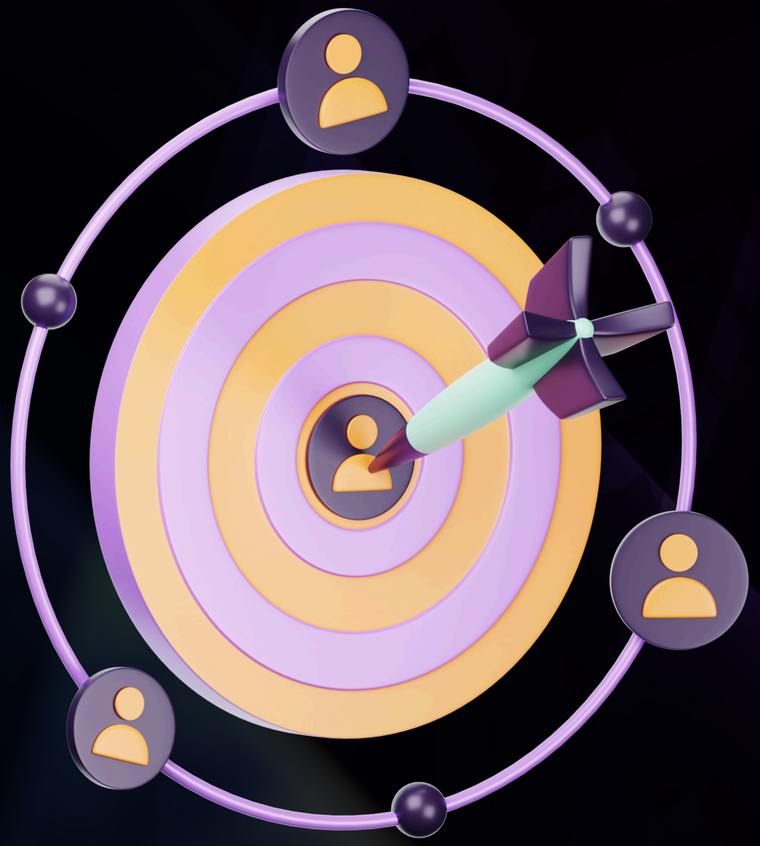
PLAIN-LANGUAGE RESULTS

Deliver verdicts in simple, non-technical language anyone can understand

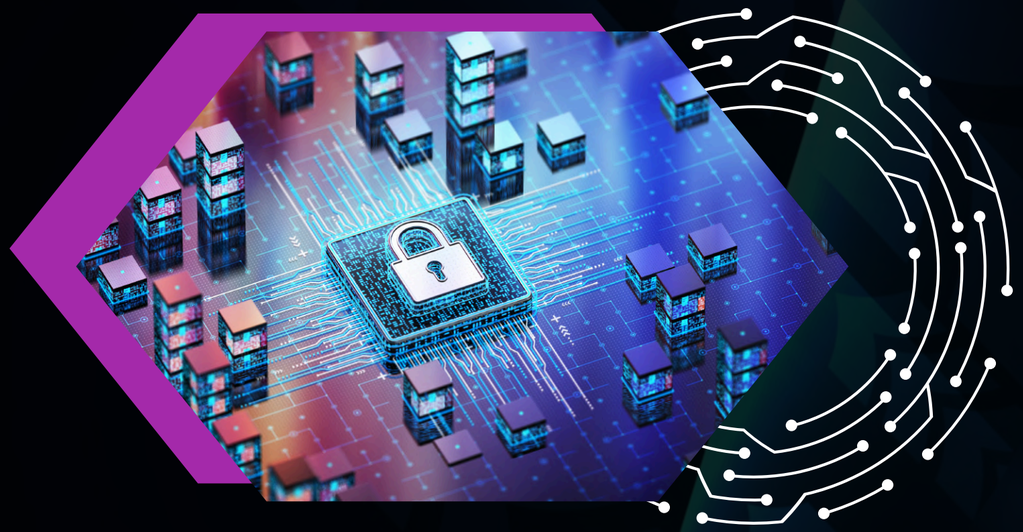
RESILIENT & GRACEFUL

Handle edge cases gracefully to ensure users are never left without guidance.

THE SOLUTION



HOW WE SOLVE THIS



WHAT THIS BOT DOES

- **Accepts real-world inputs** (messages, URLs, screenshots) via Telegram
- **Uses multi-layer detection** (Regex/OCR + VirusTotal + GLM reasoning) for higher accuracy than manual checks
- Provides clear, structured threat verdicts for fast decision-making
- Centralizes all incidents into a real-time dashboard for organizational visibility and tracking

ADVANTAGES OVER OTHER METHODS

- Eliminates manual phishing triage by automating the entire Level 1 SOC workflow, reducing ~20 minutes per incident to near-instant analysis
- Reduces SOC workload by shifting humans from manual analysis to approval-only (HITL for high-risk cases)

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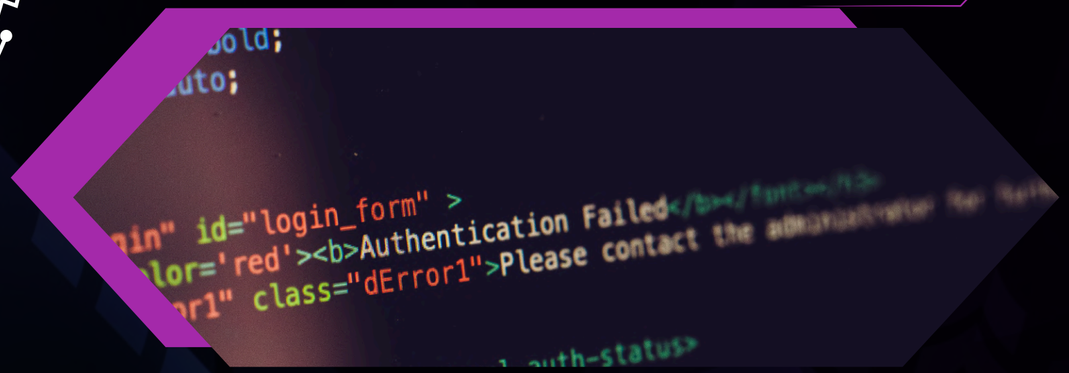
TECH OVERVIEW



Core components:

- Regex + OCR.space → data extraction
- VirusTotal API → threat intelligence
- ILMU-GLM-5.1 → structured reasoning engine

Key feature: GLM outputs strict JSON for automated downstream processing (no manual parsing)

Architecture	Single-server modular pipeline integrating multiple tools (OCR, VirusTotal, GLM, database)
Frontend	Telegram Bot (user input) + Streamlit Dashboard (monitoring & analytics)
Backend	Python orchestration layer handling API calls, logic, and pipeline flow
Database	SQLite SIEM logging for all incidents with caching mechanism

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HOW IT WORKS

User submits suspicious message/ image via Telegram

→ System extracts URLs/IPs using Regex or OCR

→ Extracted data is scanned using VirusTotal (multi-engine threat check)

→ GLM analyzes combined data and returns structured verdict (threat level + confidence + reasoning)

→ Risk Matrix determines next action (auto-response or escalation)

→ High-risk cases trigger Human-in-the-Loop approval via Telegram

→ Final result is defanged (safe) and sent back to user

→ All incidents are logged in SQLite and displayed on the real-time dashboard

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PRODUCT DEMO

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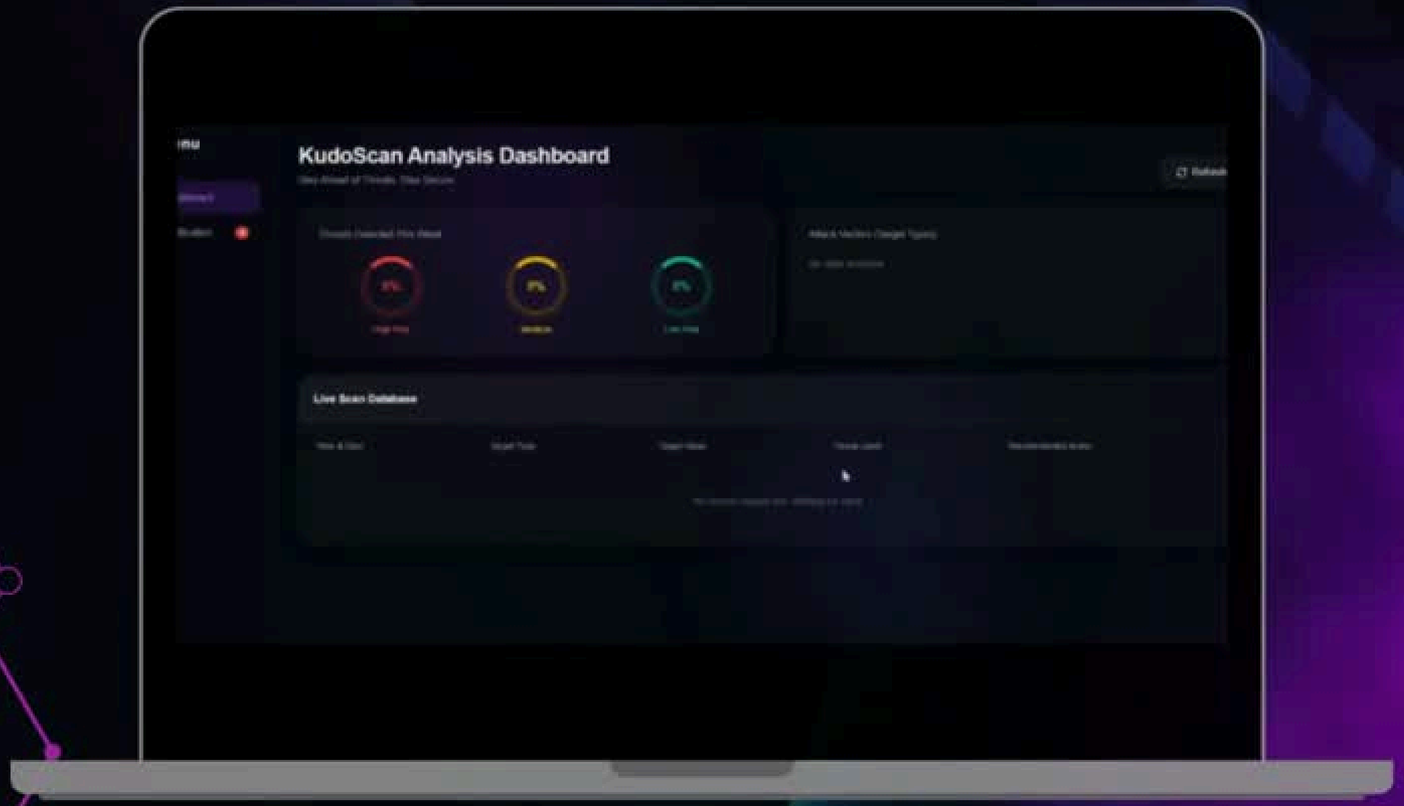

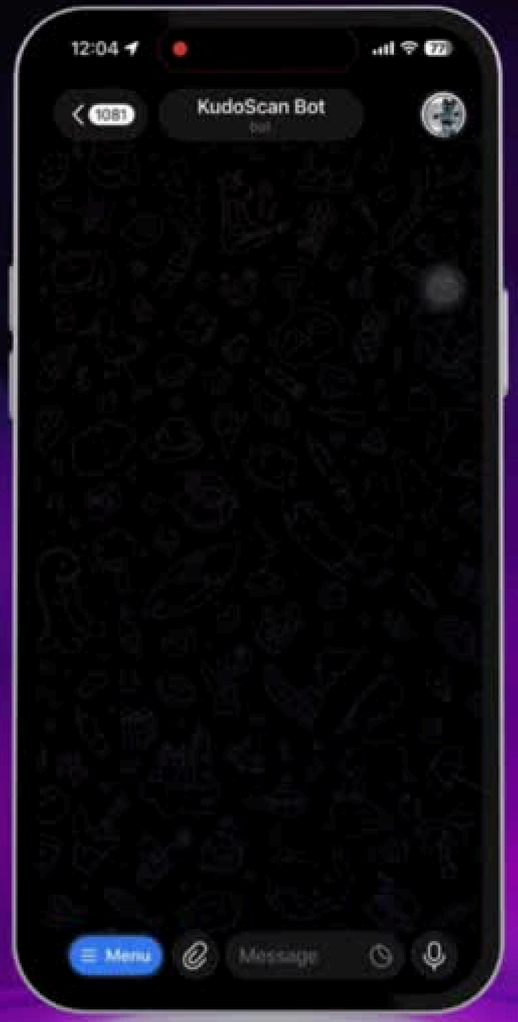

FUTURE IMPROVEMENTS

Multi-platform Support

Expand beyond Telegram to WhatsApp, Signal, and other messaging platforms.

Email Provider Integration

Direct integration with Gmail and Outlook so users don't need to manually copy-paste email content.

Multilingual Support

Deliver verdicts in the user's native language for wider accessibility.

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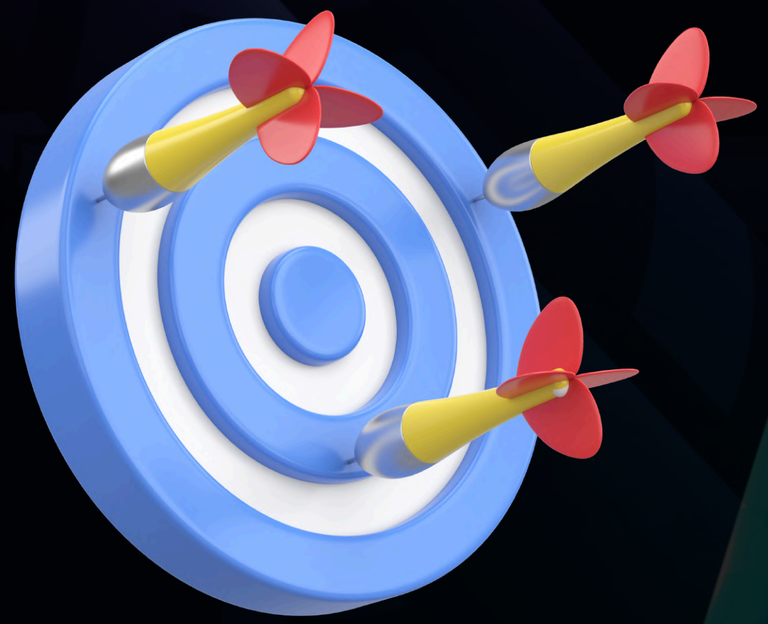

THANK YOU

FROM SUSPICIOUS TO SECURE IN SECONDS